

Tensor-Structured Attention for Classification under Variable Signal Availability

1. Purpose and working hypothesis

Test whether a structured latent representation makes attention-based multichannel classifiers less sensitive to missing channels and temporary outages. Compare a tensor-assisted model with a matched attention-only model using the same signals, features, training masks and subject splits. The working hypothesis is that shared channel–time–feature structure can help preserve task-relevant information when observations are incomplete; improvement is not assumed.

Here, a **higher-order tensor** and a **hypermatrix** refer to the same multidimensional arrays. The substantive intervention is a multilinear factor model applied to that array [4]. Simply storing inputs in a three-dimensional array is not the proposed contribution. Tensor completion has already been studied for corrupted EEG [5]; this proposal tests its relation to attention-based classification and dynamic availability.

2. Exactly where tensor methods are applied

Input representation. Extract fixed, channel-wise time–frequency features from non-overlapping windows. Arrange them as channel $\times$ temporal window $\times$ spectral feature, with a separate availability mask. Channel positions refer to a known electrode universe, so an absent channel remains identifiable. All comparison models receive the same features; unobserved entries are not treated as measured zeros.

Primary intervention — before attention. Fit a shared Tucker-2 factor model: one factor matrix describes channel structure and another describes spectral-feature structure, while the temporal-window axis remains explicit. For each incomplete recording, infer a compact core using only its observed features. Shared factors are learned from training subjects and then frozen, keeping latent component identities consistent across recordings. The core has fixed latent dimensions even when the observed channel count changes.

Attention and classification. Convert latent-component–window vectors from the core into tokens. Add component identity, temporal position and availability information, then apply the common attention classifier. Standard multi-head attention is the first integration target; repeat the main comparison with Nyströmformer-style and Performer-style attention once the tensor module is validated [1–3].

Separate reconstruction variant. Use the same core and factors to estimate missing feature vectors, preserve all observed feature values, and feed the completed channel–window tokens to attention. Supply the original mask so estimates remain distinguishable from observations. This reconstructs features, not raw EEG waveforms. It is an ablation of the latent route, not a mandatory extra stage in that route.

3. Experimental design and interpretation

Use BCI Competition IV dataset 2a: four-class motor imagery, nine participants and 22 EEG channels; exclude EOG channels [6]. Use three subject-disjoint folds with five training, one validation and three test participants, rotating the test groups, and three initialisation seeds. Fit feature normalisation and shared tensor factors on training subjects only. Freeze them before test inference.

Test 22, 16, 11 and 6 retained channels; random versus spatially grouped losses; and channels disappearing and returning during a trial. Compare static and dynamic masks at matched average availability. Mixed-availability training uses 22, 16 and 6 channels with random static masks and contiguous outages; reserve 11-channel inputs, spatial-loss patterns and separate combination banks for evaluation. Keep at least one observed channel per window.

The core comparison is **attention-only versus tensor-core → attention** under identical mixed-availability training. Start with standard attention, then repeat that pair with the two approximation methods. Also compare full-input-only versus mixed-availability training for the standard-attention pair. This separates representation effects from dropout-training effects.

With standard attention, add the feature-completion route and an ordinary masked linear bottleneck with the same latent output shape and a comparable parameter budget. These controls test whether any gain comes from multilinear structure rather than merely reducing token count or adding capacity. For completion, include a simple training-fitted imputation baseline. Keep rank and regularisation searches small and validation-only.

Primary outcomes are balanced accuracy, macro-F1, full-input degradation and paired robustness gains, reported per subject with subject-level uncertainty. Reconstruction error on artificially hidden features is secondary: lower error need not improve classification. Do not fit tensor factors jointly on training and test recordings or use hidden test features as reconstruction inputs.

Mathematical appendix

A1. Observed tensor and mask

For one recording, $C_{\max}$ is the known channel universe, P the number of temporal windows and F the number of fixed spectral features. The function f is shared with the attention-only controls. An observation mask M is broadcast over features:

$$\mathcal{X} \in \mathbb{R}^{C_{\max} \times P \times F}, \quad \mathcal{X}_{cp} = f(x_{cp}), \quad \mathcal{W}_{cpf} = M_{cp}.$$

Only entries with $M_{cp} = 1$ are available to the model. The dense notation is bookkeeping for a known channel set, not evidence that unavailable values were observed. The time–frequency transform is computed within available windows; it must not draw on hidden raw samples.

A2. Shared Tucker-2 representation

Compress channel and feature modes, while retaining the time-window mode [4]:

$$U \in \mathbb{R}^{C_{\max} \times R_C}, \quad V \in \mathbb{R}^{F \times R_F}, \quad G \in \mathbb{R}^{R_C \times P \times R_F}.$$

$$\hat{\mathcal{X}}_{cpf} = \sum_{r=1}^{R_C} \sum_{s=1}^{R_F} U_{cr} G_{rps} V_{fs}, \quad \hat{X}_p = U G_p V^T.$$

G_p is the $R_C \times R_F$ slice of the core at window p . U and V are shared across recordings; a different core is inferred for each recording. This is a Tucker-2 model with an uncompressed temporal axis. Fix small candidate ranks, for example R_C in $\{4, 8\}$ and R_F in $\{4, 8\}$, subject to feature dimensions, and select them on validation subjects.

A3. Inference from available channels

Let $O_p = \{c : M_{cp} = 1\}$. With shared factors fixed, estimate each core slice by regularised least squares:

$$\hat{G}_p = \underset{G \in \mathbb{R}^{R_C \times R_F}}{\operatorname{argmin}} \frac{\|X_{O_p, p, :} - U_{O_p, :} G V^T\|_F^2}{|O_p| F} + \lambda_G \|G\|_F^2.$$

The numerator uses only observed channel rows. $\lambda_G > 0$ makes the quadratic problem strictly convex in G for fixed factors; it does not make the estimate statistically reliable when information is insufficient. No hidden feature or test label is used. The initial implementation uses independent window-wise solves, without future-window smoothing.

A4. Learning the shared factors

Let ℓ_p denote the complete regularised expression in A3. Learn U , V and training-recording cores by minimising the following training-only objective, with unit-norm columns in U and V to control scaling:

$$\min_{U, V, \{G^{(n)}\}} \sum_{n \in \mathcal{D}_{\text{train}}} \sum_{p=1}^P \ell_p \left(G_p^{(n)}; U, V, X^{(n)}, M^{(n)} \right).$$

Use alternating optimisation, then freeze U and V for classifier training and evaluation. At inference, solve only for the current recording's core. Independent decompositions fitted from scratch to each test recording are not interchangeable because component ordering and scale can differ.

Identifiability limit. A channel absent from every training recording cannot acquire an identifiable free factor row from these data alone. Even a previously observed channel may not be recoverable after severe information loss. Shared factors provide a learned prior, not a guarantee; fixed latent size does not imply invariance to missing information.

Mathematical appendix — continued

A5. Latent tokens and attention

Map each core vector to d dimensions; a_r identifies latent component r , τ_p marks its original time, and W_M embeds the observed-channel mask. Give the same mask information to the matched controls:

$$z_{rp} = W_G \hat{G}_{rp} + a_r + \tau_p + W_M M_{:p}, \quad Z \in \mathbb{R}^{(R_C P) \times d}.$$

$$H = \text{Block}_{\text{att}}(Z), \quad \hat{y} = \text{softmax} \left(W_o \frac{1}{N_Z} \sum_{i=1}^{N_Z} H_i + b_o \right).$$

$N_Z = R_C P$ for the latent route. $\text{Block}_{\text{att}}$ uses standard softmax attention, a Nyström landmark approximation or a Performer positive-feature approximation [1–3], with the same residual/feed-forward structure. The missingness enters through core inference and the mask embedding; attention no longer receives a variable number of channel tokens.

A6. Feature-completion route and training

$$\tilde{\mathcal{X}} = \mathcal{W} \odot \mathcal{X} + (1 - \mathcal{W}) \odot \hat{\mathcal{X}}.$$

The symbol $\odot$ denotes element-wise multiplication. In implementation, take values only where the mask selects them; do not multiply uninitialised missing entries by zero. Tokenise each completed feature vector with channel/time identity and its original availability flag. Tensor factors remain frozen; train the attention classifier with:

$$\mathcal{L}_{\text{cls}} = \mathbb{E}_{(X, Y), M \sim p_{\text{train}}} [-\log \hat{y}_Y(X, M)] + \lambda_{\theta} \|\theta\|_2^2.$$

Only classifier and token-adapter parameters belong to θ . Reconstruction fitting and supervised classifier training are separate stages in the initial study. End-to-end fine-tuning is not needed for the primary comparison.

A7. Classification and reconstruction outcomes

Let $S(q, g)$ be subject-averaged balanced accuracy at retained fraction q and loss pattern g . Use the same mask bank for both members of every comparison. K_q is the number of tested retained fractions; ε is a small stabiliser:

$$G(q, g) = S_{\text{tensor}}(q, g) - S_{\text{baseline}}(q, g), \quad R_g = \frac{1}{K_q} \sum_{k=1}^{K_q} S(q_k, g).$$

$$\text{NRMSE}_{\text{hid}} = \frac{\|\mathcal{W}_{\text{hid}} \odot (\hat{\mathcal{X}} - \mathcal{X})\|_F}{\|\mathcal{W}_{\text{hid}} \odot \mathcal{X}\|_F + \varepsilon}.$$

$\mathcal{W}_{\text{hid}}$ selects only originally observed features artificially withheld for evaluation. NRMSE is defined in feature space. Report paired gain G , full-input score and degradation, mean robustness R_g , and uncertainty across subjects. Test targets are used for scoring only, never for fitting or core inference.